

AC Optimal Power Flow in Rectangular Coordinates with Augmented Lagrangian Formulation

1 Lagrange Rectangular-Coordinate ACOPF Model

1.1 Sets and Parameters

- $\mathcal{N}$: set of buses, indexed by i, j .
- $\mathcal{G} \subseteq \mathcal{N}$: set of generator buses.
- $\mathcal{L}$: set of transmission lines (i, j) .
- P_{Di}, Q_{Di} : active and reactive load at bus i .
- $Y_{ij} = G_{ij} + jB_{ij}$: (i, j) entry of the bus admittance matrix. Here G_{ij} and B_{ij} are the real and imaginary parts, respectively.
- For each line $(i, j) \in \mathcal{L}$, $y_{ij} = g_{ij} + jb_{ij}$ is the series admittance of the branch.
- $V_i^{\min}, V_i^{\max}$: lower and upper limits on voltage magnitude at bus i .
- $P_{Gi}^{\min}, P_{Gi}^{\max}$: active power limits of generator $i \in \mathcal{G}$.
- $Q_{Gi}^{\min}, Q_{Gi}^{\max}$: reactive power limits of generator $i \in \mathcal{G}$.
- $S_{ij}^{\text{total, sq}}, S_{ij, \max}^{\text{total, sq}}$: Total power squared and its limit of branch $(i, j) \in \mathcal{L}$.
- $C_i(P_{Gi})$: generation cost function at generator $i \in \mathcal{G}$, typically a convex quadratic or piecewise-linear function of P_{Gi} .

1.2 Decision Variables

The decision (optimization) variables of the rectangular-coordinate ACOPF are

$$\begin{array}{ll}
 P_{Gi}, Q_{Gi}, & \forall i \in \mathcal{G}, \\
 V_i^R, V_i^I, \mathfrak{V}_i & \forall i \in \mathcal{N}, \\
 P_{ij}, Q_{ij}, S_{ij}^{\text{total, sq}} & \forall (i, j) \in \mathcal{L},
 \end{array}$$

where

$$V_i = V_i^R + jV_i^I$$

is the complex bus voltage at bus i written in rectangular coordinates, and $P_{ij} + jQ_{ij}$ is the complex power flow from bus i to bus j .

1.3 Explicit Rectangular-Coordinate ACOPF Formulation

The ACOPF in rectangular coordinates can be written as:

$$\min_{\substack{P_G, Q_G, \\ V^R, V^I, \\ P_{ij}, Q_{ij}}} \sum_{i \in \mathcal{G}} C_i(P_{Gi}) \quad (1)$$

$$\text{s.t.} \quad P_{Gi} - P_{Di} = V_i^R \sum_{j \in \mathcal{N}} (G_{ij} V_j^R - B_{ij} V_j^I) + V_i^I \sum_{j \in \mathcal{N}} (G_{ij} V_j^I + B_{ij} V_j^R), \quad \forall i \in \mathcal{N}, \quad (2)$$

$$Q_{Gi} - Q_{Di} = V_i^I \sum_{j \in \mathcal{N}} (G_{ij} V_j^R - B_{ij} V_j^I) - V_i^R \sum_{j \in \mathcal{N}} (G_{ij} V_j^I + B_{ij} V_j^R), \quad \forall i \in \mathcal{N}, \quad (3)$$

$$P_{ij} = V_i^R [g_{ij}(V_i^R - V_j^R) - b_{ij}(V_i^I - V_j^I)] + V_i^I [g_{ij}(V_i^I - V_j^I) + b_{ij}(V_i^R - V_j^R)], \quad \forall (i, j) \in \mathcal{L}, \quad (4)$$

$$Q_{ij} = V_i^I [g_{ij}(V_i^R - V_j^R) - b_{ij}(V_i^I - V_j^I)] - V_i^R [g_{ij}(V_i^I - V_j^I) + b_{ij}(V_i^R - V_j^R)], \quad \forall (i, j) \in \mathcal{L}, \quad (5)$$

$$\mathfrak{V}_i = (V_i^R)^2 + (V_i^I)^2, \quad \forall i \in \mathcal{N}, \quad (6)$$

$$S_{ij}^{\text{total, sq}} = P_{ij}^2 + Q_{ij}^2, \quad \forall (i, j) \in \mathcal{L}. \quad (7)$$

$$V_{\text{ref}}^I = 0, \quad (8)$$

$$P_{Gi}^{\min} \leq P_{Gi} \leq P_{Gi}^{\max}, \quad \forall i \in \mathcal{G}, \quad (9)$$

$$Q_{Gi}^{\min} \leq Q_{Gi} \leq Q_{Gi}^{\max}, \quad \forall i \in \mathcal{G}, \quad (10)$$

$$\mathfrak{V}_i^{\min} \leq \mathfrak{V}_i \leq \mathfrak{V}_i^{\max}, \quad \forall i \in \mathcal{N}, \quad (11)$$

$$0 \leq S_{ij}^{\text{total, sq}} \leq S_{ij, \max}^{\text{total, sq}}, \quad \forall (i, j) \in \mathcal{L}. \quad (12)$$

The model (1)–(7)

is the ACOPF model in a rectangular coordinate system where all equations are explicitly expanded,

where all decision variables are

$$\{P_{Gi}, Q_{Gi}\}_{i \in \mathcal{G}}, \quad \{V_i^R, V_i^I, \mathfrak{V}_i\}_{i \in \mathcal{N}}, \quad \{P_{ij}, Q_{ij}, S_{ij}^{\text{total, sq}}\}_{(i, j) \in \mathcal{L}}.$$

2 Lagrangian Reformulation (Rectangular Form)

Let

$$\mathbf{x} = (P_G, Q_G, V^R, V^I, \mathfrak{V}, P_{ij}, Q_{ij}, S^{\text{total, sq}})$$

collect all primal variables, where $\mathfrak{V} = (\mathfrak{V}_i)_{i \in \mathcal{N}}$ and $S^{\text{total, sq}} = (S_{ij}^{\text{total, sq}})_{(i, j) \in \mathcal{L}}$.

We now treat all equality constraints in the model, including bus power balance, branch-flow definitions, auxiliary-variable definitions, and the reference-bus condition, as constraints to be dualized in a classical Lagrangian (without quadratic penalty).

Bus power balance constraints. For each bus $i \in \mathcal{N}$,

$$c_i^P(\mathbf{x}) := P_{Gi} - P_{Di} - \left[V_i^R \sum_{j \in \mathcal{N}} (G_{ij} V_j^R - B_{ij} V_j^I) + V_i^I \sum_{j \in \mathcal{N}} (G_{ij} V_j^I + B_{ij} V_j^R) \right] = 0, \quad (13)$$

$$c_i^Q(\mathbf{x}) := Q_{Gi} - Q_{Di} - \left[V_i^I \sum_{j \in \mathcal{N}} (G_{ij} V_j^R - B_{ij} V_j^I) - V_i^R \sum_{j \in \mathcal{N}} (G_{ij} V_j^I + B_{ij} V_j^R) \right] = 0. \quad (14)$$

Branch power-flow definition constraints. For each line $(i, j) \in \mathcal{L}$,

$$c_{ij}^{P, \text{flow}}(\mathbf{x}) := P_{ij} - \left[V_i^R (g_{ij}(V_i^R - V_j^R) - b_{ij}(V_i^I - V_j^I)) + V_i^I (g_{ij}(V_i^I - V_j^I) + b_{ij}(V_i^R - V_j^R)) \right] = 0, \quad (15)$$

$$c_{ij}^{Q, \text{flow}}(\mathbf{x}) := Q_{ij} - \left[V_i^I (g_{ij}(V_i^R - V_j^R) - b_{ij}(V_i^I - V_j^I)) - V_i^R (g_{ij}(V_i^I - V_j^I) + b_{ij}(V_i^R - V_j^R)) \right] = 0. \quad (16)$$

Voltage-magnitude auxiliary-variable constraints. For each bus $i \in \mathcal{N}$,

$$c_i^{\mathfrak{V}}(\mathbf{x}) := \mathfrak{V}_i - [(V_i^R)^2 + (V_i^I)^2] = 0. \quad (17)$$

Branch total power (squared) auxiliary-variable constraints. For each line $(i, j) \in \mathcal{L}$,

$$c_{ij}^S(\mathbf{x}) := S_{ij}^{\text{total, sq}} - (P_{ij}^2 + Q_{ij}^2) = 0. \quad (18)$$

Reference-bus constraint. For the reference bus,

$$c^{\text{ref}}(\mathbf{x}) := V_{\text{ref}}^I = 0. \quad (19)$$

Collecting all these constraints, we define

$$h(\mathbf{x}) = \begin{bmatrix} \{c_i^P(\mathbf{x})\}_{i \in \mathcal{N}} \\ \{c_i^Q(\mathbf{x})\}_{i \in \mathcal{N}} \\ \{c_{ij}^{P, \text{flow}}(\mathbf{x})\}_{(i,j) \in \mathcal{L}} \\ \{c_{ij}^{Q, \text{flow}}(\mathbf{x})\}_{(i,j) \in \mathcal{L}} \\ \{c_i^{\mathfrak{V}}(\mathbf{x})\}_{i \in \mathcal{N}} \\ \{c_{ij}^S(\mathbf{x})\}_{(i,j) \in \mathcal{L}} \\ c^{\text{ref}}(\mathbf{x}) \end{bmatrix}. \quad (20)$$

Let the corresponding Lagrange multipliers be λ , stacked in the same order as the components of $h(\mathbf{x})$.

The generation cost is

$$f(\mathbf{x}) = \sum_{i \in \mathcal{G}} C_i(P_{Gi}). \quad (21)$$

2.1 Classical Lagrangian Function (Quadratic in $\mathbf{x}$)

With all equality constraints collected in $h(\mathbf{x})$, the (classical) Lagrangian function is

$$\mathcal{L}(\mathbf{x}, \lambda) = f(\mathbf{x}) + \lambda^\top h(\mathbf{x}) + \left(\frac{\gamma^\top}{2} \|x - \hat{x}\|^2 \right), \quad (22)$$

where $\hat{x}$ is the solution from the previous round (which is optional), λ are treated as dual variables (parameters) and $\mathbf{x}$ are primal variables. Because $f(\mathbf{x})$ is quadratic in the generator outputs and each component of $h(\mathbf{x})$ is at most quadratic in the voltage and flow variables, $\mathcal{L}(\mathbf{x}, \lambda)$ is at most quadratic in $\mathbf{x}$ for fixed λ . This is compatible with QHD solvers that require a quadratic Hamiltonian.

2.2 Linearized Lagrangian Formulation

Although the classical Lagrangian in (22) is quadratic in the objective variables, the equality constraints $h(\mathbf{x})$ contain nonlinear AC power flow relations involving bilinear voltage products. To obtain a structure compatible with quadratic Hamiltonian solvers, we linearize the constraint functions around a reference operating point $\hat{\mathbf{x}}$.

Using a first-order Taylor expansion, the constraint vector can be approximated as

$$h(\mathbf{x}) \approx h(\hat{\mathbf{x}}) + J(\hat{\mathbf{x}})(\mathbf{x} - \hat{\mathbf{x}}), \quad (23)$$

where

$$J(\mathbf{x}) = \frac{\partial h(\mathbf{x})}{\partial \mathbf{x}} \quad (24)$$

is the Jacobian matrix of the constraint functions.

Substituting this approximation into the Lagrangian yields the linearized Lagrangian

$$\mathcal{L}_{\text{lin}}(\mathbf{x}, \lambda) = f(\mathbf{x}) + \lambda^\top [h(\hat{\mathbf{x}}) + J(\hat{\mathbf{x}})(\mathbf{x} - \hat{\mathbf{x}})] + \frac{\gamma}{2} \|\mathbf{x} - \hat{\mathbf{x}}\|^2. \quad (25)$$

Because the constraint approximation is affine in $\mathbf{x}$, the resulting Lagrangian remains quadratic in the decision variables. Expanding the terms gives

$$\mathcal{L}_{\text{lin}}(\mathbf{x}, \lambda) = f(\mathbf{x}) + \lambda^\top J(\hat{\mathbf{x}})\mathbf{x} + \frac{\gamma}{2} \|\mathbf{x} - \hat{\mathbf{x}}\|^2 + \lambda^\top (h(\hat{\mathbf{x}}) - J(\hat{\mathbf{x}})\hat{\mathbf{x}}), \quad (26)$$

where the final term is constant with respect to $\mathbf{x}$. Therefore the optimization problem reduces to minimizing a quadratic function of $\mathbf{x}$, which can be directly mapped to a quadratic Hamiltonian suitable for QHD-based solvers.

$$x = \begin{bmatrix} P_{G1} \\ Q_{G1} \\ V_{R1} \\ V_{R2} \\ V_{I1} \\ V_{I2} \\ V_{sq,1} \\ V_{sq,2} \\ P_{12} \\ Q_{12} \\ S_{12}^2 \end{bmatrix}, \quad h(x) = \begin{bmatrix} c_1^P \\ c_2^P \\ c_1^Q \\ c_2^Q \\ c_{12}^{P, \text{flow}} \\ c_{12}^{Q, \text{flow}} \\ c_1^{V^2} \\ c_2^{V^2} \\ c_{12}^S \\ c^{\text{ref}} \end{bmatrix}, \quad J(x) = \frac{\partial h(x)}{\partial x}.$$

$$y_{12} = g_{12} + j b_{12} = \frac{1}{r_{12} + j x_{12}}.$$

$$J(x) = \begin{bmatrix} 1 & 0 & -g_{12}(2V_{R1} - V_{R2}) - b_{12}V_{I2} & g_{12}V_{R1} + b_{12}V_{I1} & -g_{12}(2V_{I1} - V_{I2}) + b_{12}V_{R2} & g_{12}V_{I1} - b_{12}V_{R1} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & g_{12}V_{R2} + b_{12}V_{I2} & -g_{12}(2V_{R2} - V_{R1}) - b_{12}V_{I1} & g_{12}V_{I2} - b_{12}V_{R2} & -g_{12}(2V_{I2} - V_{I1}) + b_{12}V_{R1} & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 2b_{12}V_{R1} - b_{12}V_{R2} - g_{12}V_{I2} & -b_{12}V_{R1} + g_{12}V_{I1} & 2b_{12}V_{I1} - b_{12}V_{I2} + g_{12}V_{R2} & -b_{12}V_{I1} - g_{12}V_{R1} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -b_{12}V_{R2} + g_{12}V_{I2} & 2b_{12}V_{R2} - b_{12}V_{R1} - g_{12}V_{I1} & -b_{12}V_{I2} - g_{12}V_{R2} & 2b_{12}V_{I2} - b_{12}V_{I1} + g_{12}V_{R1} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -g_{12}(2V_{R1} - V_{R2}) - b_{12}V_{I2} & g_{12}V_{R1} + b_{12}V_{I1} & -g_{12}(2V_{I1} - V_{I2}) + b_{12}V_{R2} & g_{12}V_{I1} - b_{12}V_{R1} & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 2b_{12}V_{R1} - b_{12}V_{R2} - g_{12}V_{I2} & -b_{12}V_{R1} + g_{12}V_{I1} & 2b_{12}V_{I1} - b_{12}V_{I2} + g_{12}V_{R2} & -b_{12}V_{I1} - g_{12}V_{R1} & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & -2V_{R1} & 0 & -2V_{I1} & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -2V_{R2} & 0 & -2V_{I2} & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -2P_{12} & -2Q_{12} & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} P_{G1} \\ Q_{G1} \\ V_{R1} \\ V_{R2} \\ V_{I1} \\ V_{I2} \\ V_{sq,1} \\ V_{sq,2} \\ P_{12} \\ Q_{12} \\ S_{12}^2 \end{bmatrix} \quad (27)$$

after the plug-in of the G, B, g, b , We have

$$J(x) = \begin{bmatrix} 1 & 0 & -1.2437372875(2V_{R1} - V_{R2}) + 5.0960209212V_{I2} & 1.2437372875V_{R1} - 5.0960209212V_{I1} & -1.2437372875(2V_{I1} - V_{I2}) - 5.0960209212V_{R2} & 1.2437372875V_{I1} + 5.0960209212V_{R1} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1.2437372875V_{R2} - 5.0960209212V_{I2} & -1.2437372875(2V_{R2} - V_{R1}) + 5.0960209212V_{I1} & 1.2437372875V_{I2} + 5.0960209212V_{R2} & -1.2437372875(2V_{I2} - V_{I1}) - 5.0960209212V_{R1} & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & -10.1920418424V_{R1} + 5.0960209212V_{R2} - 1.2437372875V_{I2} & 5.0960209212V_{R1} + 1.2437372875V_{I1} & -10.1920418424V_{I1} + 5.0960209212V_{I2} + 1.2437372875V_{R2} & 5.0960209212V_{I1} - 1.2437372875V_{R1} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 5.0960209212V_{R2} + 1.2437372875V_{I2} & -10.1920418424V_{R2} + 5.0960209212V_{R1} - 1.2437372875V_{I1} & 5.0960209212V_{I2} - 1.2437372875V_{R2} & -10.1920418424V_{I2} + 5.0960209212V_{I1} + 1.2437372875V_{R1} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -1.2437372875(2V_{R1} - V_{R2}) + 5.0960209212V_{I2} & 1.2437372875V_{R1} - 5.0960209212V_{I1} & -1.2437372875(2V_{I1} - V_{I2}) - 5.0960209212V_{R2} & 1.2437372875V_{I1} + 5.0960209212V_{R1} & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & -10.1920418424V_{R1} + 5.0960209212V_{R2} - 1.2437372875V_{I2} & 5.0960209212V_{R1} + 1.2437372875V_{I1} & -10.1920418424V_{I1} + 5.0960209212V_{I2} + 1.2437372875V_{R2} & 5.0960209212V_{I1} - 1.2437372875V_{R1} & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & -2V_{R1} & 0 & -2V_{I1} & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -2V_{R2} & 0 & -2V_{I2} & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -2P_{12} & -2Q_{12} & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} P_{G1} \\ Q_{G1} \\ V_{R1} \\ V_{R2} \\ V_{I1} \\ V_{I2} \\ V_{sq,1} \\ V_{sq,2} \\ P_{12} \\ Q_{12} \\ S_{12}^2 \end{bmatrix} \quad (28)$$

In Simple

$$J(x) =$$

$$\begin{bmatrix} 1 & 0 & -1.24(2V_{R1} - V_{R2}) + 5.10V_{I2} & 1.24V_{R1} - 5.10V_{I1} & -1.24(2V_{I1} - V_{I2}) - 5.10V_{R2} & 1.24V_{I1} + 5.10V_{R1} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1.24V_{R2} - 5.10V_{I2} & -1.24(2V_{R2} - V_{R1}) + 5.10V_{I1} & 1.24V_{I2} + 5.10V_{R2} & -1.24(2V_{I2} - V_{I1}) - 5.10V_{R1} & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & -10.20V_{R1} + 5.10V_{R2} - 1.24V_{I2} & 5.10V_{R1} + 1.24V_{I1} & -10.20V_{I1} + 5.10V_{I2} + 1.24V_{R2} & 5.10V_{I1} - 1.24V_{R1} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 5.10V_{R2} + 1.24V_{I2} & -10.20V_{R2} + 5.10V_{R1} - 1.24V_{I1} & 5.10V_{I2} - 1.24V_{R2} & -10.20V_{I2} + 5.10V_{I1} + 1.24V_{R1} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -1.24(2V_{R1} - V_{R2}) + 5.10V_{I2} & 1.24V_{R1} - 5.10V_{I1} & -1.24(2V_{I1} - V_{I2}) - 5.10V_{R2} & 1.24V_{I1} + 5.10V_{R1} & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & -10.20V_{R1} + 5.10V_{R2} - 1.24V_{I2} & 5.10V_{R1} + 1.24V_{I1} & -10.20V_{I1} + 5.10V_{I2} + 1.24V_{R2} & 5.10V_{I1} - 1.24V_{R1} & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & -2V_{R1} & 0 & -2V_{I1} & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -2V_{R2} & 0 & -2V_{I2} & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -2P_{12} & -2Q_{12} & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} P_{G1} \\ Q_{G1} \\ V_{R1} \\ V_{R2} \\ V_{I1} \\ V_{I2} \\ V_{sq,1} \\ V_{sq,2} \\ P_{12} \\ Q_{12} \\ S_{12}^2 \end{bmatrix} \quad (29)$$

after plug-in $V_R = 1$ and $V_I = 0$ We have

$$J(x) = \begin{bmatrix} 1 & 0 & -1.24 & 1.24 & -5.10 & 5.10 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1.24 & -1.24 & 5.10 & -5.10 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & -5.10 & 5.10 & 1.24 & -1.24 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 5.10 & -5.10 & -1.24 & 1.24 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -1.24 & 1.24 & -5.10 & 5.10 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & -5.10 & 5.10 & 1.24 & -1.24 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & -2 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -2 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -2P_{12} & -2Q_{12} & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} P_{G1} \\ Q_{G1} \\ V_{R1} \\ V_{R2} \\ V_{I1} \\ V_{I2} \\ V_{sq,1} \\ V_{sq,2} \\ P_{12} \\ Q_{12} \\ S_{12}^2 \end{bmatrix} \quad (30)$$

3 Box-Constrained Feasible Set and Primal–Dual Iterations

After moving all equality constraints into the Lagrangian, the remaining constraints are simple bound (box) constraints on the variables. We denote the feasible set by

$$\mathcal{X} = \left\{ \mathbf{x} : P_{Gi}^{\min} \leq P_{Gi} \leq P_{Gi}^{\max}, \quad \forall i \in \mathcal{G}, \right. \quad (31)$$

$$Q_{Gi}^{\min} \leq Q_{Gi} \leq Q_{Gi}^{\max}, \quad \forall i \in \mathcal{G}, \quad (32)$$

$$\mathfrak{V}_i^{\min} \leq \mathfrak{V}_i \leq \mathfrak{V}_i^{\max}, \quad \forall i \in \mathcal{N}, \quad (33)$$

$$0 \leq S_{ij}^{\text{total, sq}} \leq S_{ij, \max}^{\text{total, sq}}, \quad \forall (i, j) \in \mathcal{L}, \quad (34)$$

$$\underline{V}_i^R \leq V_i^R \leq \overline{V}_i^R, \quad \underline{V}_i^I \leq V_i^I \leq \overline{V}_i^I, \quad \forall i \in \mathcal{N}, \quad (35)$$

$$\underline{P}_{ij} \leq P_{ij} \leq \overline{P}_{ij}, \quad \underline{Q}_{ij} \leq Q_{ij} \leq \overline{Q}_{ij}, \quad \forall (i, j) \in \mathcal{L} \}, \quad (36)$$

where the bounds $\underline{V}_i^R, \overline{V}_i^R, \underline{V}_i^I, \overline{V}_i^I, \underline{P}_{ij}, \overline{P}_{ij}, \underline{Q}_{ij}, \overline{Q}_{ij}$ are chosen according to engineering limits (e.g., derived from feasible voltage and power ranges).

In compact notation, this can be written as

$$\mathcal{X} = \{\mathbf{x} : \underline{\mathbf{x}} \leq \mathbf{x} \leq \overline{\mathbf{x}}\}.$$

A primal–dual (dual ascent) scheme then solves the sequence of box-constrained subproblems

$$\mathbf{x}^{k+1} = \arg \min_{\mathbf{x} \in \mathcal{X}} \mathcal{L}(\mathbf{x}, \lambda^k), \quad (37)$$

$$\lambda^{k+1} = \lambda^k + \alpha_k h(\mathbf{x}^{k+1}), \quad (38)$$

where k is the iteration index and $\alpha_k > 0$ is a dual stepsize (e.g., a scalar or a diagonal scaling matrix). At each iteration, for fixed λ^k the inner problem (37) is a nonconvex quadratic (in $\mathbf{x}$) box-constrained optimization problem, which can be approximately solved by a QHD-based solver. The dual variables are then updated by (38) to reduce the constraint residual $h(\mathbf{x}^{k+1})$.